Aditya Prasanna Mogare

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EDUCATION

M.S., Computer Science | ***University of Southern California*** **(GPA: 3.70/4.0)** **Los Angeles, CA Aug 2024 - May 2026**

B.Eng, Computer Engineering | ***University of Mumbai*** **(GPA: 9.63/10)** **Mumbai, India Aug 2020 - May 2024**

EXPERIENCE

Easley Dunn Productions | ***Software Engineer (Contract)*** **Jul 2026 - Present**

- Built a Python LLM agent with LangGraph and LangChain to draft and validate JSON game levels so designers could keep Daily Challenges on schedule without new hires, cutting creation time from 3 hours to 35 minutes.
- Implemented retrieval-augmented generation (RAG) using pgvector on PostgreSQL to show the agent previously approved layouts instead of costly retraining, raising first-try rule compliance from 58% to 86%.
- Wrote C# Unity editor tools and a JSON-driven prefab pipeline with a versioned ScriptableObject schema to replace duplicated scene files, shrinking the project by 93% (3 GB to 200 MB).

INTERDEPENDENT Studios | ***Software & AI Automation Engineer*** **May 2025 - Aug 2025**

- Architected an event-driven Python Flask webhook pipeline using the OpenAI API to grade PDF submissions on a 6-category rubric so a small team could run open calls at scale, scoring 5,000+ entries with zero manual review.
- Integrated Webflow forms, Stripe Identity, and Mailgun email through REST APIs and webhooks to give writers one submit-and-verify flow, a blueprint used in a grant application that received $10M in New York State funding.
- Hardened the platform using webhook retries, idempotency checks, and state tracking on Dockerized Flask services on AWS EC2 and Amazon RDS, so no submission was lost or scored twice.

University of Southern California | ***Graduate Research Assistant*** **Jan 2025 - Jun 2025**

- Trained a ResNet50 CNN in PyTorch, run in Docker on Kubernetes over AWS EC2 and S3, to automate early screening for diabetic retinopathy, a diabetes-related eye disease, reaching 85% validation accuracy.

HiringTek and Kansastek | ***AI Engineer Co-op*** **Aug 2023 - Dec 2023**

- Developed an interview question-bank service using Python Flask, the OpenAI GPT-3 API, MongoDB, and an Angular form to generate oral and coding questions by domain, topic, and level, with an LLM duplicate check and a model-written answer for every question.
- Automated first-pass interview review using Google Cloud Speech-to-Text, Praat speech rate, GingerIt grammar, NLP answer scoring, and OpenCV and Keras CNN face analysis to score answers and flag integrity issues.

Software Engineer Co-op (same company) **Aug 2022 - Jul 2023**

- Launched recorded technical, screening, and HR interviews using WebRTC audio, video, and screen capture, with a Google Cloud Storage media service that split .webm files into mp4 and wav via FFmpeg for scoring.
- Ranked applicants against each role using Python parsing of must-have and nice-to-have skills from job descriptions and resumes, plus an in-house code runner for Python, C, C++, and SQL, so recruiters reviewed the strongest candidates first.

Threat Prism | ***AI Engineer Co-op*** **Dec 2021 - Aug 2022**

- Engineered a Python NLP classification pipeline using NLTK, scikit-learn TF-IDF, and fine-tuned BERT from Hugging Face to replace manual news sorting, tagging 15,000+ articles into 10 categories at 87% accuracy.

Diginique TechLabs | ***Full Stack Developer Co-op*** **Jun 2021 - Dec 2021**

- Created responsive educational websites using React, Typescript, JavaScript, HTML CSS, and a Node.js and Express REST API on MongoDB to give students course content on any device, reaching 400+ students.

PROJECTS

RelayAI – Computer-Use Agent | Demo **Sep 2026**

- TypeScript, Node.js, Playwright, LLM Agents, Docker, GitHub Actions
- Constructed a TypeScript computer-use agent with Playwright that used an LLM once to learn a banking workflow, then replayed it deterministically to make automation affordable, cutting runs from 6 minutes to 4 seconds.

StudyAI – Classroom Memory Agent | Demo **Aug 2026**

- AWS Lambda, API Gateway, EventBridge, Bedrock, CockroachDB, Vector Search, Discord
- Composed a serverless, event-driven AWS system using Lambda, API Gateway, EventBridge, and Bedrock to match student Discord questions to syllabus topics with vector search, giving TAs weekly weak-topic reports.

DevLite – Developer Environment Doctor | Demo **Sep 2026**

- C++20, CMake, GoogleTest, Homebrew, GitHub Actions, macOS
- Programmed a C++20 CLI with an extensible probe-and-rule design to diagnose PATH, Git, Python, and pip issues in one command, verified by 50 GoogleTest tests in GitHub Actions CI/CD and shipped via CMake and Homebrew.

SKILLS

Languages: Python, C++, C, C#, TypeScript, JavaScript, SQL, HTML/CSS
AI / ML: LLM Agents, RAG, Prompt Engineering, LLM Evaluation, pgvector, LangGraph, LangChain, OpenAI, Bedrock, PyTorch, Keras, CNNs, NLP, BERT, Hugging Face, scikit-learn, NLTK, OpenCV, Tesseract OCR, Praat, GingerIt, VoiceIt
Backend & Data: REST APIs, Microservices, Webhooks, Flask, Node.js, Express, PostgreSQL, CockroachDB, MongoDB, Mailgun
Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, API Gateway, EventBridge), GCP (Speech-to-Text, Storage), Docker, Kubernetes, Linux, Nginx, Gunicorn, CI/CD
Frontend & Tools: React, Angular, WebRTC, Socket.IO, FFmpeg, Unity, Playwright, GoogleTest, CMake, Homebrew, Git, Stripe